Counting Fruits with State-of-the-Art Multi-View-Stereo Approaches

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Abstract

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Chapter 1

Introduction

Feeding a growing world population is a major global challenge, and projections estimate that total food demand will rise substantially over the coming decades [7]. Precision agriculture responds to this by using advanced technology and data analysis to maximize crop yields while reducing waste and environmental impact [17]. Within precision agriculture, counting the fruit on a tree is an important building block: it provides the yield estimates that growers rely on to plan the harvest and allocate labor, storage, and other resources [27]. Because performing the count manually across many trees is slow and labor-intensive, automating it from images is a valuable capability.

Among the ways to automate fruit counting, reconstructing the tree in 3D from a set of RGB images is particularly attractive, for two reasons. First, lifting the count into a shared three-dimensional representation lets each fruit be counted exactly once, instead of being counted repeatedly across the many overlapping images in which it appears. Second, capturing the necessary data requires only an ordinary color camera, of the kind built into any consumer device, rather than the specialized and more expensive depth-sensing equipment, such as LiDAR scanners or calibrated stereo rigs, on which other 3D-capture methods depend. This makes the approach inexpensive and easy to deploy in the field.

Recent work such as FruitNeRF [25] shows that image-based 3D fruit counting is feasible by training a Neural Radiance Field [26] per scene and counting the fruit in the resulting volumetric representation. This approach is effective, but it requires a time-consuming optimization for every new scene. In this work, we instead investigate whether a feed-forward multi-view stereo model, VGGT [33], can replace this per-scene training. Rather than optimizing a representation for each scene, VGGT predicts a 3D reconstruction in a single forward pass. This promises a significant reduction in runtime, but raises the central question of whether the resulting geometry is accurate enough for reliable counting.

Throughout this thesis, we focus on apples. Their nearly spherical shape allows for a simple geometric model, and FruitNeRF provides a real-world dataset of apple trees with manually counted ground truth [25], which enables a direct comparison with the optimization-based approach.

1.1 Challenges

Counting fruit reliably from images of a tree faces a number of difficulties, some inherent to the trees themselves and some arising from the way they must be captured.

The trees are difficult subjects in their own right. Fruit are frequently occluded, both by foliage

and by other fruit, so that any single image captures only part of the crop. Their appearance changes with the viewing angle and with lighting conditions such as direct sunlight or shadow, which makes the same fruit look different when seen from different viewpoints. In addition, the branches, leaves and fruit of a tree canopy form a geometrically complex, cluttered structure that is hard to reconstruct consistently.

The way the tree is captured adds further difficulties. To limit occlusion and avoid missing fruit, a tree should be observed from all sides. The number of images that can be used per tree is bounded, however, by the memory available to the reconstruction model (see Section 3.1.5). To still achieve full coverage, the cameras must therefore be placed sparsely around the tree, spanning a full 360-degree range. As a consequence, neighboring images are taken from widely spaced viewpoints and often show markedly different regions of the tree, with little visual content in common. This makes it hard to establish *correspondences* between images, that is, to determine which parts of one image show the same physical points as another. The problem is compounded by the scene not being perfectly static during capture: wind moves branches, leaves, and fruit between one image and the next, so that even points that are visible in both images no longer align, further degrading the cross-view correspondences on which the later counting steps rely. These unreliable correspondences are the root cause of the counting errors this thesis addresses: depending on the situation, the same apple may be split into several counts or several apples merged into one. How these errors arise in our pipeline, and how we counteract them, is discussed in Chapter 4.

1.2 Our Approach

To count the fruit in a scene, our method maps semantic information, in the form of 2D instance masks obtained with SAM3 [3], across views by means of the 3D representation predicted by VGGT. The association of masks across views follows the point-map-based principle introduced by CCGS [16]: pixel correspondences between views are established through the predicted point maps, and instance masks are matched across views by solving an assignment problem that allows partial matches. We adopt this association step, transfer it from the image-pair setting of its origin to the multi-view reconstruction produced by VGGT, and build a new counting pipeline on top of it. In this pipeline, the pairwise mask matches are aggregated into multi-view instances through a graph formulation, and the fruit are counted by combining these associated instances with a subsequent clustering step.

1.3 Contributions

This thesis makes the following contributions. We investigate whether the feed-forward multi-view stereo model VGGT can replace per-scene optimization for image-based fruit counting, and we assess whether its predicted geometry is accurate enough for this task. Building on the point-map-based mask association introduced by CCGS [16], we develop a fruit-counting pipeline that extends this association with a graph-based aggregation of instances across many 360-degree views, sphere fitting as an apple model, and a final clustering step to obtain the count. We further propose an iterative refinement of this pipeline that detects and resolves over-merged instances, thereby addressing the opposing split-and-merge errors that arise under sparse, widely spaced views. Finally, we evaluate both

the base pipeline and the refined pipeline on data

Chapter 2

Related Work

2.1 Fruit Counting and Yield Estimation

Methods for counting fruit differ mainly in where the counting takes place: in the individual image, across a sequence of images, or in a three-dimensional representation of the scene. These stages can be read as increasingly robust answers to the core difficulty of counting across views, namely avoiding that the same fruit is counted more than once.

The earliest and simplest methods operate purely in the image plane, separating the task into detecting fruit in each image and counting the detections [12]. Detection has moved from handcrafted color and shape features to deep neural networks [27], and modern detectors such as YOLO reach high accuracy on individual images [1]. Because these methods count on a per-frame basis, however, they have no notion of which detections in different images correspond to the same fruit, so a fruit visible in several frames is counted several times.

To resolve this, a second line of work tracks fruit across consecutive frames. Liu et al. [22] proposed the first complete pipeline of this kind, using Structure-from-Motion (SfM) to link detections across frames and correct the resulting double counts, and recent pipelines pair modern detectors with dedicated tracking modules for the same purpose [4]. In later work, Liu et al. [23] matched detections between frames with the Hungarian algorithm on top of a semantic SfM reconstruction, using a single camera. What these methods share is a reliance on densely sampled video: consecutive frames differ only slightly, so detections can be tracked through strong frame-to-frame correspondences, and the association is typically driven by 2D image-space cues such as bounding-box overlap or optical flow between adjacent views. Our setting is the opposite. To cover a tree from all sides within a limited view budget, the cameras are placed sparsely around a full 360-degree range, so neighboring views often show markedly different regions of the tree and share little image-space overlap. Frame-to-frame tracking in the image plane is therefore not applicable, and the association must instead be established in 3D, through the reconstructed geometry rather than through 2D overlap. We adopt the Hungarian algorithm for this association as well, but apply it to correspondences derived from the 3D reconstruction rather than 2D bounding-box overlap between adjacent frames.

A third line of work removes double counting structurally by moving the count into three dimensions, so that each fruit exists once in a shared spatial representation. Häni et al. [12] reconstructed rows of trees from both sides and merged the counts from the combined 3D models. FruitNeRF [25] takes

this idea furthest by integrating the counting directly into a neural scene representation, unifying reconstruction and segmentation that earlier methods had kept separate. Its drawback is that the neural representation must be optimized for each scene, which is computationally expensive; this cost is the immediate starting point of the present work. Newer methods reduce it by replacing the neural field with faster representations, such as the 3D Gaussian splatting used by FruitGaussian [15] and CountingFruit [21], but they remain tied to a per-scene optimization. A different way to avoid this optimization is to associate instance masks across views through a 3D representation rather than to optimize one; since this is the strategy of the present work, we treat it separately in the next section.

2.2 Multi-View Instance Association and 3D Segmentation

Associating 2D instance masks across many views into consistent 3D instances is a problem closely related to fruit counting, and one that general-purpose 3D scene-understanding methods address independently of any counting application. What distinguishes these methods is the criterion by which they decide that two masks seen in different views depict the same object.

The most direct criterion is geometric overlap in an explicit reconstruction. Gené-Molá et al. [11] segment fruit with Mask R-CNN, project the 2D masks into an SfM point cloud, associate detections by their overlap in 3D, and discard false detections with a support vector machine. The association is accurate but inherits the cost of the underlying SfM reconstruction, which the authors note is computationally intensive and unsuitable for real-time use.

A second criterion replaces this geometric overlap with cross-view agreement. MaskClustering [36] builds a global graph whose nodes are the 2D masks and weights each edge by a *view consensus rate*, the fraction of masks from other views that contain both endpoints; instances then emerge from iteratively clustering the masks that agree most strongly. The decision is thereby made globally, over all views at once, rather than only between neighboring frames, but it rests on how masks co-occur across views rather than on explicit 3D correspondences.

The criterion our method builds on, introduced by CCGS [16], derives the association directly from dense point maps. Pixel correspondences between views are found by minimizing the Euclidean distance between the predicted point maps; the overlap between two masks is redefined in terms of these correspondences; and the resulting matching cost is solved with the Hungarian algorithm, with partial matches permitted. CCGS uses the associated masks to build a compact segmentation field for 3D Gaussian splatting, and it operates on image pairs from a DUSt3R reconstruction. We take this point-map-based association and, instead, apply it to the joint multi-view reconstruction produced by VGGT, using it as the first stage of our counting pipeline.

Chapter 3

Fundamentals

3.1 VGGT

The central component of the reconstruction stage in this work is VGGT (Visual Geometry Grounded Transformer), a feed-forward transformer introduced by Wang et al. [33] and awarded the Best Paper Award at CVPR 2025. In this section, we first contextualize VGGT within the broader field of 3D reconstruction and then examine its input data, internal pipeline, and outputs. Finally, we discuss the properties that make it a suitable foundation for our counting method.

3.1.1 Context

Reconstructing the three-dimensional structure of a scene from a set of two-dimensional images is a long-standing problem in computer vision [13]. The classical solution is a multi-stage pipeline consisting of Structure-from-Motion (SfM) and Multi-View Stereo (MVS). In the implementation by Schönberger and Frahm [31], SfM estimates the camera poses as well as a sparsely populated point cloud by extracting and matching handcrafted features across the images, followed by an iterative optimization process known as bundle adjustment. A subsequent MVS step then densifies this reconstruction. While these methods are accurate and remain widely used, they rely on iterative optimization that is computationally expensive and can take several minutes to hours per scene to complete.

More recently, learning-based approaches have replaced parts of this pipeline with neural networks. Neural Radiance Fields (NeRFs) [26] represent a scene as a volumetric radiance function encoded in the weights of a neural network, while 3D Gaussian Splatting [18] models it as a collection of colored Gaussian distributions. Both produce high-quality reconstructions but require separate optimization for each individual scene, which again incurs a substantial runtime cost per reconstruction.

A different line of work aims to eliminate per-scene optimization entirely by predicting geometry with a neural network. DUSt3R [34] pioneered this *feed-forward* principle by regressing a dense point map from an image pair in a single network pass, followed by MASt3R [20], which added feature matching. However, these methods operate on image pairs and still require a global alignment step to fuse the pairwise predictions into a consistent scene. VGGT [33] belongs to this feed-forward family, but generalizes it from pairs to arbitrary sets of images: it processes one, a few, or up to hundreds of views together in a single forward pass and predicts a globally consistent reconstruction without any

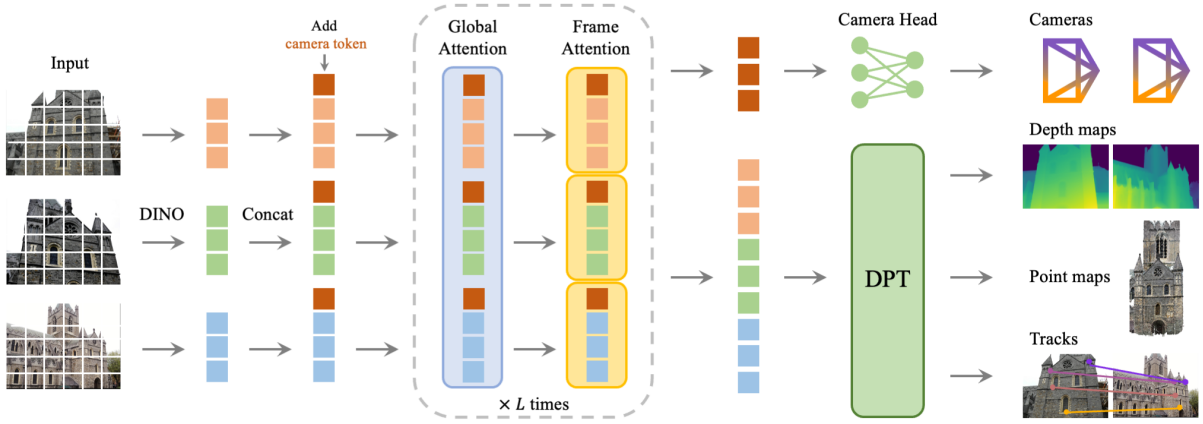


Figure 3.1: Overview of the VGGT architecture, taken from [33]. The input images are split into patch tokens by DINO and augmented with camera tokens. $L = 24$ blocks of frame-wise and global self-attention process the tokens of all images jointly, after which the camera head predicts the camera parameters and the DPT head the dense outputs.

subsequent optimization. This combination of speed and multi-view consistency motivates its use in this work, where fast reconstruction is a central goal.

3.1.2 Input Data

The input for VGGT is a sequence of N RGB images $(I_i)_{i=1}^N$ with $I_i \in \mathbb{R}^{3\times H\times W}$, which capture the same static scene. Unlike the classical SfM pipeline, no camera parameters, calibration data, or prior pose information need to be provided: the model derives all geometric quantities directly from the image content. The number of views N is flexible and can range from a single image to several hundred, with the only practical limitation being the available GPU memory, since all views are processed jointly [33].

3.1.3 Pipeline

Internally, VGGT maps the input sequence through three conceptual stages to a set of dense 3D predictions: tokenization, alternating attention, and task-specific prediction heads. An overview is shown in Figure 3.1.

Tokenization. Each input image I_i is first divided into a grid of non-overlapping image patches. Each patch is then encoded into a *token*, a fixed-length vector that represents the local content of that patch and serves as the basic unit on which the network operates [8]. In VGGT, this patch encoding is performed by a pretrained DINOv2 backbone [29]. An image is thus converted into a set of tokens, each of which carries the local appearance and structural information of its region. For each image, the patch tokens are augmented with additional learnable tokens: a *camera token*, which aggregates the information needed to predict the camera parameters of this image, as well as a set of *register tokens*, which serve as auxiliary memory during the attention calculation and are discarded before the final prediction.

Alternating Attention. The core of the network is a transformer [32] that processes all tokens of all images jointly. It consists of 24 attention blocks, each of which applies frame-wise and global self-attention in turn. In *frame-wise* self-attention, tokens attend only to other tokens of the same image, which preserves and refines the local structure of each image. In *global* self-attention, all tokens attend to one another across all images, thereby exchanging information between views and establishing the correspondences that link the individual images into a single, coherent scene. By alternating between these two modes, the network balances local detail with global multi-view consistency. This alternating attention design is the central architectural mechanism of VGGT; the authors show that it is superior to both purely global attention and cross-attention alternatives [33].

Prediction Heads. After the attention blocks, the refined tokens are passed to specialized prediction heads. A dedicated *camera head* processes the camera tokens to regress the intrinsic and extrinsic camera parameters of each view. A *DPT head* [30], a Dense Prediction Transformer, upsamples the image tokens back to the input resolution to generate the dense, pixel-wise outputs, namely depth maps, point maps, tracking features, and the corresponding confidence maps.

3.1.4 Output

For each input image I_i, VGGT predicts the following quantities:

- camera extrinsics and intrinsics of the view,
- a dense depth map together with a per-pixel confidence map,
- a dense point map together with a per-pixel confidence map, and
- dense feature maps used for tracking pixels across views.

The point map is the output most relevant to this work. It assigns a 3D scene coordinate $P_i(y) \in \mathbb{R}^3$ to every pixel y of an image, thereby representing a dense, pixel-wise 3D reconstruction. Crucially, the point maps of all images are expressed in a single, common coordinate system: they are defined in the reference frame of the first camera, which serves as the world origin [33]. Points observed in different views therefore already lie in a common 3D space without requiring an explicit alignment or registration step. This shared coordinate frame is the property our method builds upon, as it allows 2D instance masks from different views to be related to one another purely through their 3D coordinates.

Each dense prediction is accompanied by a confidence map that estimates the pixel-wise reliability of the reconstruction. These confidence values provide a signal for identifying and filtering out unreliable regions, such as poorly observed or geometrically ambiguous areas [33].

3.1.5 Advantages and Limitations

The main advantage of VGGT for our application is its speed. Since the entire reconstruction is generated in a single forward pass, a scene can be reconstructed in a matter of seconds, whereas the SfM-based and scene-specific optimization approaches discussed in Section 3.1.1 take minutes to hours. At the same time, according to the authors, VGGT matches or exceeds the accuracy of these optimization-based alternatives across multiple 3D tasks [33], and does so without any post-processing. Another practical

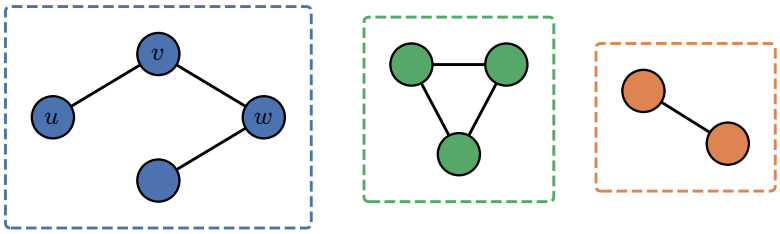


Figure 3.2: A graph with three connected components (dashed). The nodes u and w are not adjacent, but they belong to the same component because they are connected by a path via v.

advantage is that it operates directly on uncalibrated images and outputs camera parameters, depth, and point maps together, so that no separate calibration or pose estimation step is required.

The fundamental limitation is that all input views are processed collectively via global self-attention, whose memory requirements increase with the number of tokens and, consequently, with the number of input views. The authors note that a naive implementation can become very memory intensive when dealing with a large number of tokens, so that in practice, the available GPU memory limits the number of images that can be reconstructed together [33]. This constraint is directly relevant to our scenario, in which a fruit tree must be captured from all sides: The limited view budget forces a sparse distribution of cameras around the scene, which in turn weakens the correspondences between neighboring views. A number of follow-up works address this scaling limit, for example by processing long sequences in segments, as in VGGT-Long [6] and VGGT-SLAM [24], or by scaling up to thousands of images as in Fast3R [37]; Pi3 [35] additionally eliminates the dependence on a fixed reference view. These extensions are beyond the scope of this work, which uses the original VGGT model.

3.2 Graph Theory

This section introduces the graph theory concepts used in Chapter 4: connected components and the Union-Find data structure, which computes them efficiently.

3.2.1 Basic Concepts

A graph $G = (V, E)$ consists of a set of vertices (or nodes) V and a set of edges $E \subseteq V \times V$, where each edge connects a pair of vertices. Two vertices are said to be *adjacent* if they are connected by an edge. A *path* is a sequence of vertices in which every consecutive pair of vertices is connected by an edge. A *connected component* is a maximal subset of vertices in which every pair of vertices is connected by a path and which is not connected to any vertex outside the subset. Intuitively, a connected component is a group of nodes that are all directly or indirectly connected to one another, while being separated from the rest of the graph.

Connectivity is transitive: If node u is connected to v via a path and v is connected to w, then u is also connected to w. Thus, a connected component unites nodes that were never directly connected by an edge, as long as they are connected by a chain of edges. Figure 3.2 illustrates this for a small graph.

3.2.2 Union-Find

Computing the connected components of a graph via a disjoint-set data structure is a standard technique [5]. We use the *Union-Find* data structure, also known as the disjoint-set data structure, which manages a partition of the nodes into disjoint sets and efficiently supports two operations: `find`, which returns a representative element for the set containing a given node, and `union`, which merges the two sets containing two given nodes into one.

Each set is stored as a tree, in which every node points to a parent node, and the root of the tree serves as the representative of the set. The `find` operation follows the parent pointers from a node up to the root. Two nodes belong to the same set if and only if the `find` operation returns the same root for both. The `union` operation joins two sets by attaching the root of one tree as a child of the root of the other.

To compute the connected components within a graph, the Union-Find structure is initialized with a singleton set per node, and `union` is called for the two endpoints of every edge. After all edges have been processed, the sets of the Union-Find structure are exactly the connected components of the graph [5].

A naive implementation of `find` can degenerate when the trees become tall, since each call then traverses a long chain of parent pointers. To avoid this, *path compression* is applied: During a `find` operation, every node visited on the path to the root is immediately made to point directly to the root [5]. This makes repeated lookups on the same set progressively cheaper and keeps the extraction of connected components efficient even for large graphs.

3.3 Hungarian Matching

The assignment problem asks for an optimal one-to-one pairing between the elements of two sets. This section introduces the problem, the Hungarian algorithm [19, 28], which solves it exactly, and partial matching, which allows elements to remain unassigned.

3.3.1 The Assignment Problem

The assignment problem is a classical combinatorial optimization problem. In its original formulation [19], a set of n workers is to be assigned n jobs, given a numerical rating for the performance of each worker on each job, such that the total rating of the chosen assignment is optimal and no worker is assigned more than one job. More generally, the elements of two sets are paired, and each candidate pair carries a cost.

Let $A = \{1, \dots, n_A\}$ and $B = \{1, \dots, n_B\}$ be two sets with elements $a \in A$ and $b \in B$. For every pair (a, b), a cost $C_{ab} \geq 0$ expresses how poorly the two elements fit together, where a lower cost indicates a better pair. These costs form a cost matrix $C \in \mathbb{R}^{n_A \times n_B}$.

In its standard form, the linear sum assignment problem assumes an equal number of elements on both sides and assigns each element exactly once. Describing an assignment by binary variables $x_{ab} \in \{0, 1\}$, where $x_{ab} = 1$ means that a is matched to b, it is written with equality constraints as $\sum_b x_{ab} = 1$ and $\sum_a x_{ab} = 1$, so that every element receives exactly one partner [2]. If the two sets differ in size, this is not possible. Assuming $n_A \leq n_B$ without loss of generality, the rectangular form of the

problem assigns every element of the smaller set exactly once and every element of the larger set at most once:

$$\begin{aligned} \min_{x} \quad & \sum_{a} \sum_{b} C_{ab}\, x_{ab} \\ \text{s.t.} \quad & \sum_{b} x_{ab} = 1 \quad \forall a, \\ & \sum_{a} x_{ab} \leq 1 \quad \forall b, \\ & x_{ab} \in \{0, 1\}. \end{aligned} \tag{3.1}$$

Because elements of A are only ever paired with elements of B, and never with elements of their own set, the underlying structure is a *bipartite* graph: its nodes split into the two disjoint sets, and every candidate match connects one set to the other, with no edges inside either set. The cost matrix C is the weighted adjacency matrix of this bipartite graph, and the assignment problem is the search for a minimum-cost bipartite matching.

Because of this formulation, $n_B - n_A$ elements of the larger set always remain unassigned. Conversely, every element of the smaller set is assigned a partner, even if none of the candidates fit well. Such assignments are subsequently removed through partial matching (Section 3.3.3).

3.3.2 The Hungarian Algorithm

The difficulty of the assignment problem is that the number of possible assignments grows factorially with the number of elements: for two sets with n elements each, there are $n!$ distinct one-to-one assignments [28], so enumerating them and picking the cheapest is infeasible for all but the smallest cases. A tempting shortcut is a greedy strategy in which each element is assigned to its individually best partner. Such a choice is locally optimal but can be globally suboptimal: two elements of A might both prefer the same element of B, which violates the one-to-one requirement, and resolving the conflict greedily can force a worse overall assignment than necessary.

The Hungarian algorithm avoids both problems. Introduced by Kuhn [19], it is named for the underlying results of the Hungarian mathematicians D. König and E. Egerváry on which it is based, and it computes a globally optimal assignment without enumerating all possibilities. Munkres [28] later gave a variant of the method together with a proof that it terminates after a number of operations bounded by a polynomial in n, rather than the factorial cost of exhaustive search; because of this joint contribution the method is also known as the Kuhn-Munkres algorithm. It therefore solves the problem in Equation (3.1) exactly and efficiently.

3.3.3 Partial Matching

The optimal assignment minimizes the total cost, but it does not by itself guarantee that the matched pairs are meaningful: the algorithm will continue to pair two poorly fitting elements if doing so reduces the total cost. In a *partial matching*, elements may instead remain unassigned when no suitable partner exists. By first solving the assignment problem and then discarding all assigned pairs whose cost exceeds a threshold, we obtain such a matching. The remaining pairs are part of an optimal assignment and, in addition, individually acceptable.

3.4 RANSAC

Fitting a geometric model to a set of measured points is a common task in computer vision. Whenever such measurements originate from imperfect detectors or reconstructions, they contain not only small measurement noise but also *gross errors*: points that are simply wrong rather than slightly inaccurate. A conventional estimator such as least squares offers no protection against these. As an averaging technique, it relies on the assumption that any deviation from the model is a small measurement error and that such deviations balance each other out over a sufficiently large number of points [10]. Gross errors clearly violate this assumption, and even a single severely misplaced point can distort the entire fit. To robustly estimate a model under such conditions, we use RANSAC (Random Sample Consensus).

RANSAC was introduced by Fischler and Bolles [10] and reverses the logic of conventional smoothing: Instead of fitting the model to all points and then attempting to remove those that do not match, it fits the model to a small sample and then measures how many of the remaining points match it. Each iteration proceeds in three steps. First, it draws a *minimal sample*, the smallest number of points that suffices to instantiate the model, and computes the model from this sample alone. Second, it tests the instantiated model against all other points and collects every point that lies within a fixed error tolerance of it into the *consensus set* for this sample; points outside the tolerance are considered outliers. Third, the size of the consensus set is recorded as the value for this sample. These steps are repeated over a specified number of iterations, and the model with the largest consensus set is retained. Optionally, this final model is refitted to all points in its consensus set to refine the estimate [10].

The robustness of the method stems from this construction. Since the model is selected by consensus among many points rather than by minimizing a residual across all points, gross outliers that would derail an averaging estimator are excluded from the consensus set and therefore do not influence the retained model. Figure 3.3 illustrates this on the simple case of fitting a circle to a set of 2D points: three points determine a candidate circle, and the points within the tolerance band form its consensus set, while the outliers that would bias a least-squares fit are left out.

3.5 Clustering

Clustering divides a set of points into groups, or clusters, so that points that are far apart belong to different clusters, while points that are close to each other belong to the same cluster. First, we will discuss density-based clustering and then turn to the agglomerative method, which is better suited for spatially compact clusters.

3.5.1 Density-Based Clustering

A widely used density-based clustering algorithm is DBSCAN, introduced by Ester et al. [9]. Its central idea is that clusters are regions of high point density separated from one another by regions of low density. Density is evaluated based on a neighborhood radius ε and a minimum number *MinPts*: A point is a *core point* if at least *MinPts* points lie within a distance of ε from it [9]. For two-dimensional data, the authors fix *MinPts* to a constant value, so that only ε remains to be chosen [9].

Clusters are then formed from core points based on a concept of *density-reachability*: A point is density-reachable from a core point if it can be connected to it by a chain of points in which each

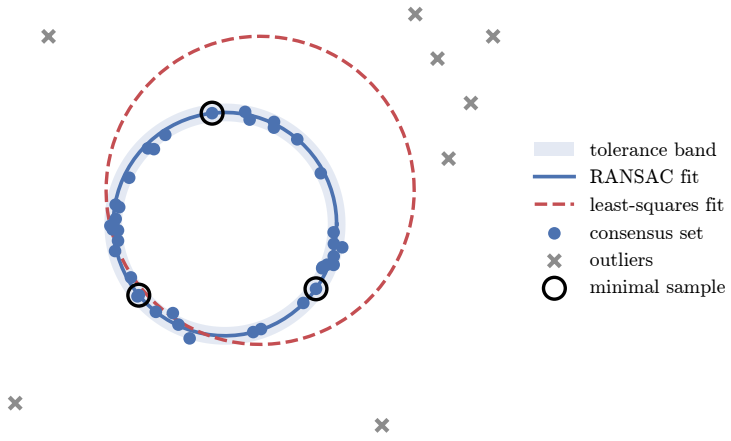


Figure 3.3: RANSAC fitting a circle to 2D points with outliers. Three points (minimal sample) determine a candidate circle, and the points within the tolerance band form its consensus set. The least-squares fit to all points (dashed) is pulled toward the outliers.

successive point lies within ε of the previous one and each point except possibly the last is itself a core point [9]. A cluster is a maximal set of density-connected points, and points that do not belong to any cluster are referred to as noise.

This chaining allows DBSCAN to discover clusters of any shape, but it can be a disadvantage if clusters are expected to be spatially compact. Since membership propagates transitively along chains of ε-close points, two dense groups are kept separate from each other only if the distance between them is greater than ε [9]. If such groups are close to one another, the points of one may lie within ε of the other, and DBSCAN then connects them via the intervening points to a single cluster. Different groups are thereby linked into a single one. Reducing ε to break such chains is not a solution: it also separates loosely connected points within a single group, thereby splitting a cluster into several. In general, there is no single radius that separates neighboring groups while simultaneously uniting the points of each individual group.

3.5.2 Agglomerative Clustering with Complete Linkage

We therefore use agglomerative hierarchical clustering, which controls the spatial extent of a cluster directly rather than via a density chain. The method begins with each point in its own cluster and repeatedly merges the two nearest clusters [14, pp. 520, 523], where the distance between two clusters is defined by a *linkage* criterion. Under *complete linkage*, this distance is the greatest distance between any pair of points, one from each cluster [14, p. 523]; two clusters are thus merged only if *all* of their points are close to one another. The merging process ends as soon as the smallest distance between the clusters exceeds a specified threshold [14, p. 521].

Since complete linkage limits the maximum distance within a cluster, it also limits the diameter that a cluster can reach and tends to produce compact clusters [14, p. 524]. It therefore cannot link a sequence of points into a group of arbitrary length. This behavior, known as *chaining* [14, p. 524], is what affects density-based clustering as described above. Choosing the threshold according to the

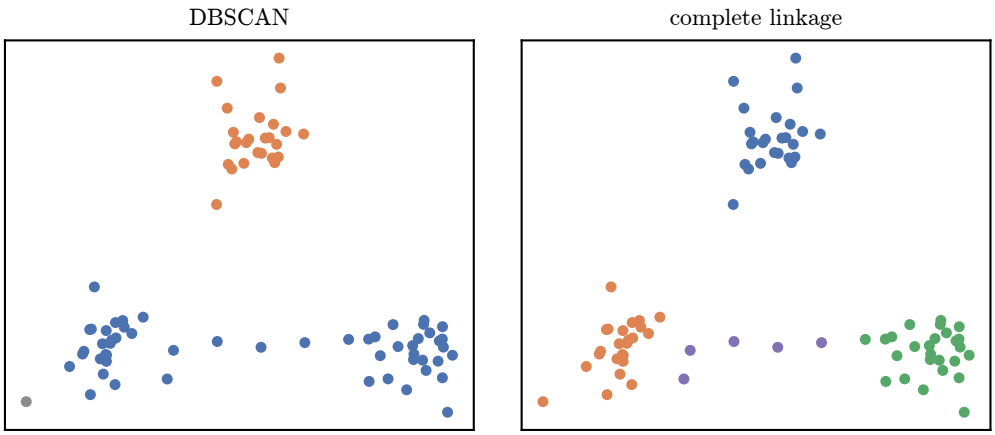


Figure 3.4: DBSCAN and agglomerative clustering with complete linkage on three compact groups, two of which are connected by a chain of points. DBSCAN links these two groups into a single cluster (left; gray: noise). Complete linkage keeps all three groups apart, and the chain forms a small cluster of its own (right).

expected size of a group makes it possible to merge the points of a group while separate groups remain distinct from one another, so that each resulting cluster corresponds exactly to one group. Figure 3.4 contrasts both methods on three compact groups, two of which are connected by a thin chain of points.

Chapter 4

Method

4.1 Overview

Our method takes as input a set of N RGB images $(I_i)_{i=1}^N$ of a tree, which were captured from different viewpoints around the tree, and returns the number of apples on the tree. As for VGGT itself (Section 3.1.2), no camera parameters are required. The method combines two complementary sources of information: VGGT provides the geometry by mapping each pixel in each view to the shared 3D coordinate system, while SAM3 [3] provides the semantics by segmenting individual apples in each view as 2D instance masks. However, the instance identifiers generated by SAM3 are meaningful only in a single view. It is not known which masks in the different views represent the same apple, and restoring these correspondences forms the core of the counting task.

We formulate this task on the graph introduced in Section 3.2.1. Its nodes are all instance masks (i, m) across all views, and the goal is to divide these into groups so that each group contains exactly those masks that represent a single physical apple. The number of groups then corresponds to the count. Two opposite types of errors can occur: a *split error* distributes the masks of one apple over several groups and leads to overcounting, while a *merge error* combines the masks of several apples in one group and leads to undercounting. Every step included in the pipeline was designed with these two errors in mind.

Figure 4.1 provides an overview of the pipeline. After determining the point maps and instance masks of all views (Section 4.2), we match the masks between each pair of views and aggregate the pairwise matches into multi-view instances (Section 4.3). Each resulting instance is checked against a geometric model of an apple, a sphere (Section 4.4), and the centers of the accepted spheres are clustered to obtain the final count (Section 4.5). Section 4.6 analyzes the errors in this pipeline, and Section 4.7 describes an iterative refinement that addresses these errors. We refer to the pipeline without this refinement as the *base pipeline* and to the pipeline including it as the *refined pipeline*. Further details on the implementation can be found in Section 4.8.

The mask association between two views, i.e., the pixel correspondences derived from the distances in the point map, as well as the matching using partial assignments, follows CCGS [16]. Aggregation over many views, the apple model, clustering, and iterative refinement are contributions of this work. To illustrate each individual step, we use a small scene as an example (Figure 4.2) throughout this chapter. It consists of seven apples lying on a table and partially separated by a cardboard partition, such that

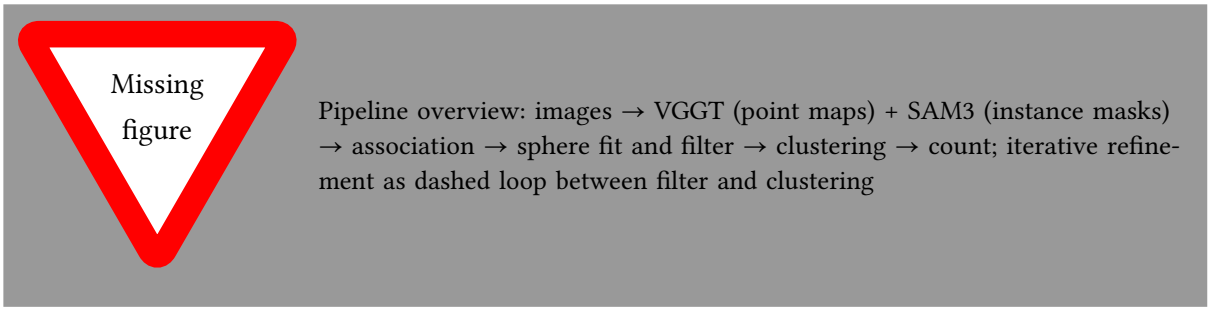


Figure 4.1: Overview of the counting pipeline.

Figure 4.2: Running example used throughout this chapter: five views of seven apples on a table, partially separated by a cardboard partition.

some apples are obscured from certain viewpoints. The scene was captured with a smartphone camera in five individual frames. Thanks to its small size, we can visualize intermediate results such as cost matrices and graphs, while the partition simulates, on a very small scale, the occlusion that makes association difficult with real trees.

4.2 Input: Reconstruction and Segmentation

4.2.1 3D Reconstruction

All N images are processed together using VGGT in a single forward pass. We directly use the output of the point map head (Section 3.1.4) as the point map P_i of view i. Wang et al. report that unprojecting the predicted depth maps using the predicted camera parameters yields more accurate points than the point map head [33]. In our setting with sparse 360-degree capture, we observe the opposite: The unprojected points from different views align less well with each other, whereas the point map head places the same surface point observed in different views almost at the same location (Chapter 5). Since the association relies entirely on spatial accuracy between views, we use the point map head. Its predictions are expressed in the coordinate system of the first view, so the property described in Section 3.1.4 is applicable. VGGT normalizes the scale of each reconstruction, so all distances and thresholds in this work are given in the units of the respective reconstruction rather than in meters.

4.2.2 Instance Segmentation

Each view is segmented independently using SAM3 [3]. Given the text prompt “apple”, SAM3 detects the instances of this concept in the image and outputs a mask for each of them. Masks whose area is smaller than a minimum area threshold are excluded to avoid small false positives. Each detected mask

receives an identifier m that is unique within its view.

SAM3 operates at the original image resolution, while the point maps are predicted at a lower VGGT input resolution. Therefore, we store the masks of each view as a single image consisting of instance identifiers and resize it to match the resolution of the point maps using nearest-neighbor interpolation. Unlike bilinear interpolation, this method never generates intermediate values that would confuse the identifiers of two neighboring instances. Thus, each pixel y of a mask corresponds to exactly one point $P_i(y)$. We denote by $\mathcal{M}_{i,m}$ the set of pixels of the instance m in view i, while $\mathcal{A}_i = \bigcup_m \mathcal{M}_{i,m}$ denotes the entire set of apple pixels in view i.

4.2.3 Outlier Removal

The predicted apple points are not error-free. In addition to minor deviations, they can contain isolated points that lie far from any apple surface, in particular at the edges of a mask, where the prediction changes between the apple and the background. Such points cause problems in two ways: they can lead to correspondences between the masks of two different apples (Section 4.3) and also enlarge the sphere fitted to the instance (Section 4.4).

Therefore, before the association, we remove the outliers from the apple points. The points $P_i(y)$ with $y \in \mathcal{A}_i$ from all views are combined into a single set of points, to which we apply radius outlier removal as implemented in Open3D [38]: a point is kept only if there are at least k_f points within its neighborhood of radius r_f. Since the set is combined from all views, a point on the apple's surface that was observed in multiple views is supported by points from other views, while an isolated point without such support is removed. The discarded points are not deleted, but rather marked as invalid in the point map so that the correspondence between pixels and points is preserved and invalid pixels are ignored in all subsequent steps. Only apple pixels are filtered, since all later steps are applied exclusively to these.

The confidence maps, which VGGT predicts in addition to the point maps (Section 3.1.4), could serve as an alternative filtering criterion. However, a large proportion of the apple pixels in our scene are assigned a low confidence level, so that setting a confidence threshold results in many valid points being excluded along with the outliers (Chapter 5). In contrast, the radius-based filter evaluates each point based on the geometry of its surroundings.

4.3 Cross-View Mask Association

This section describes how the instance masks of all views are fused into multi-view instances. The process consists of two steps. For each pair of views, the masks are matched using the point-map-based association method from CCGS [16] (Sections 4.3.1 to 4.3.3). Subsequently, the pairwise matches are aggregated over all pairs to form the connected components of the graph from Section 3.2.1 (Section 4.3.4).

CCGS associates masks between adjacent frames of a densely sampled sequence, in which consecutive views show largely the same content. We adapt it to the sparse 360-degree setting in three ways: correspondences are established only between apple pixels, they are subject to a distance threshold, and the matching is performed for all view pairs, not just adjacent ones.

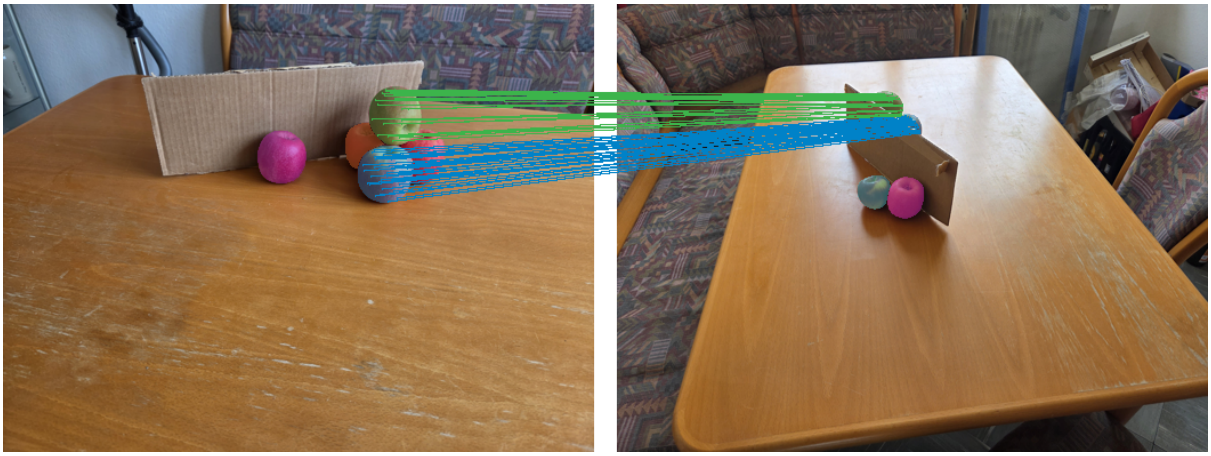


Figure 4.3: Sampled pixel correspondences from view 1 (left) to view 4 (right) of the running example, colored by instance. Apples hidden by the partition in one of the views receive hardly any correspondences.

4.3.1 Pixel Correspondence

Based on CCGS, we determine the pixel correspondence between two views i and j using their point maps: a pixel in view i corresponds to the pixel in view j whose 3D point is closest to its own [16]. Since only the apple pixels are relevant for the correspondence, we only search among these. For a pixel $y \in \mathcal{A}_i$, a candidate correspondence is given by

$$\phi_{ij}(y) = \arg\min_{y' \in \mathcal{A}_j} \|P_i(y) - P_j(y')\|_2. \tag{4.1}$$

As described in Section 4.2.3, only valid pixels are considered. In CCGS, each pixel is assigned a correspondence. This is suitable for adjacent images in which almost every point is visible in both images, which is not the case here, however. An apple that is visible in view i but obscured in view j would still be assigned to the nearest apple in view j, regardless of how far away it is, and all of its pixels would end up in the same incorrect mask. Therefore, we accept a correspondence only if

$$\|P_i(y) - P_j(\phi_{ij}(y))\|_2 < \tau_{\text{corr}}, \tag{4.2}$$

otherwise we leave y without a correspondence. The threshold τ_{corr} must account for the deviation between two predictions of the same surface point from different viewpoints, including the displacement caused by slight dynamic changes to the scene (Section 1.1). At the same time, it must remain small compared to the size of an apple. Near the point of contact between two adjacent apples, some correspondences between different apples are unavoidable, even if τ_{corr} is small. However, these affect only a few pixels, and are handled at the mask level (Section 4.3.3).

The correspondence is directed from view i to view j and is calculated once for each pair of views. Figure 4.3 illustrates this using two views from our example.

4.3.2 Mask Overlap and Matching Cost

CCGS defines the overlap of two masks in different views based on this correspondence [16]. For a mask a of view i and a mask b of view j, their intersection consists of those pixels of a whose correspondence

lies in b:

$$\mathcal{M}_{i,a} \cap_\phi \mathcal{M}_{j,b} = \{\, y \in \mathcal{M}_{i,a} \,:\, \phi_{ij}(y) \in \mathcal{M}_{j,b} \,\}, \tag{4.3}$$

where pixels without an accepted correspondence are never considered. The overlap is the size of this intersection relative to the smaller of the two masks, while the matching cost is its complement:

$$o_{ab} = \min\left(1,\ \frac{|\mathcal{M}_{i,a} \cap_\phi \mathcal{M}_{j,b}|}{\min(|\mathcal{M}_{i,a}|, |\mathcal{M}_{j,b}|)}\right), \qquad C_{ab} = 1 - o_{ab}. \tag{4.4}$$

Normalization based on the smaller mask corresponds to the CCGS method and is suitable for partial visibility. If an apple is partially obscured in one view, its smaller mask may still completely overlap with the full mask in the other view, whereas normalization based on the union of both masks would penalize the occlusion. Since multiple pixels of a may correspond to the same pixel of b, the ratio can be greater than one, which is why we restrict it to one.

4.3.3 Pairwise Matching

For a pair of views i and j with n_i and n_j masks, the costs form a matrix $C \in \mathbb{R}^{n_i \times n_j}$, and we solve the rectangular assignment problem from Equation Equation (3.1) with masks of the two views taking the roles of the sets A and B. No candidate is excluded in advance: all entries, including those with no overlap, are included in the optimization so that the solution is derived from complete information about the overlaps. Weak matches are excluded only afterward, thereby implementing the partial matching described in Section 3.3.3: An assigned pair (a, b) becomes an edge only if

$$o_{ab} > \tau_{\text{ov}}. \tag{4.5}$$

Both thresholds operate at different levels. τ_{corr} determines whether a single pixel has a corresponding pixel, while τ_{ov} determines whether the mask as a whole matches. The few correspondences at the point where two adjacent apples touch usually result in an overlap that is smaller than τ_{ov}. If this is not the case, the one-to-one assignment still prefers the true counterpart, as the following example shows.

Table 4.1 shows the overlaps between view 1 and view 4 of the example. Only two apples are visible in both views; the cardboard hides the remaining apples from view 1 in view 4 and the other way around. Thus, only two entries are close to one, and these pairs become edges between the two views. Since the rectangular assignment pairs each mask of the view with fewer masks with a partner, the two remaining masks of view 4 also receive partners in view 1, but with an overlap close to zero, which is why the threshold τ_{ov} removes them again. The table also shows why a one-to-one assignment is necessary: the mask a_2 not only overlaps with its actual corresponding mask b_3, but also overlaps by 0.21 with the mask b_2 of the neighboring apple. This value exceeds τ_{ov}, so a threshold alone would connect a_2 and b_2 and falsely merge the two apples. The assignment prevents this, since b_2 is already the best partner for a_3.

4.3.4 Aggregation into Multi-View Instances

Confirmed matches form the edges of the graph from Section 3.2.1. The Union-Find structure is initialized with a set for each mask and `union` is called for each edge. After all image pairs have been processed,

Table 4.1: Overlap o_{ab} between the masks a of view 1 (rows) and the masks b of view 4 (columns) of the running example. Bold: pairs accepted as edges; italic: pairs rejected by $\tau_{\text{ov}} = 0.2$.

	b_1	b_2	b_3	b_4
a_1	*0.00*	0.05	0.00	0.00
a_2	0.00	0.21	**1.00**	0.00
a_3	0.00	**1.00**	0.00	0.00
a_4	0.00	0.00	0.00	0.00
a_5	0.00	0.00	0.00	*0.01*

the sets form the connected components of the graph. Each component represents a hypothetical apple and is referred to below as an *instance*.

Because connectivity is transitive (Section 3.2.1), an apple observed in, for example, five views is recovered as a single component of five masks, even though no single matching step compared all five masks at once.

Unlike CCGS, which transfers labels from each individual image to the next, we match all $N(N-1)/2$ image pairs. In a sparse 360-degree capture, two images showing the same apples are not necessarily adjacent in any order, for example, two views taken at different heights on the same side of the tree. By matching all pairs, such views are directly connected to one another, rather than relying on a chain of intermediate views that breaks as soon as a connection fails.

The transitivity of the graph also has a drawback: a single incorrect edge connects two components, including all elements contained within them. Therefore, we add a constraint that is not expressed by the graph's formulation alone. An apple appears at most once in each view, so a component must not contain two masks of the same image. Before adding an edge, we check whether the components of its two nodes share the same view; if so, the edge is discarded. This excludes unions that contradict the segmentation of a single view. This constraint has two consequences. First, since the view pairs are processed in a specific order and each check depends on the edges added previously, the result may depend on this order. Second, if, for example, SAM3 splits an apple in a view into two masks because a branch obscures its central part, the constraint preserves both parts as separate components. Such fragments can be reunited through clustering.

Figure 4.4 shows the resulting graph for the running example. The 25 masks of the five views form seven connected components, one per apple. Since only 18 of the 32 accepted matches are required to connect these components, each component would stay connected even if individual matches failed, which is the advantage of matching all pairs of views.

4.4 Apple Model: Sphere Fitting and Filtering

The association returns a set of instances, each of which is a hypothesis about a single apple. Before counting them, we fit a geometric apple model to each instance. Since apples are nearly spherical, we use a sphere. The model has two purposes: it allows us to reject instances that cannot correspond to a single apple, and its center provides a compact representative of the instance for the clustering stage (Section 4.5).

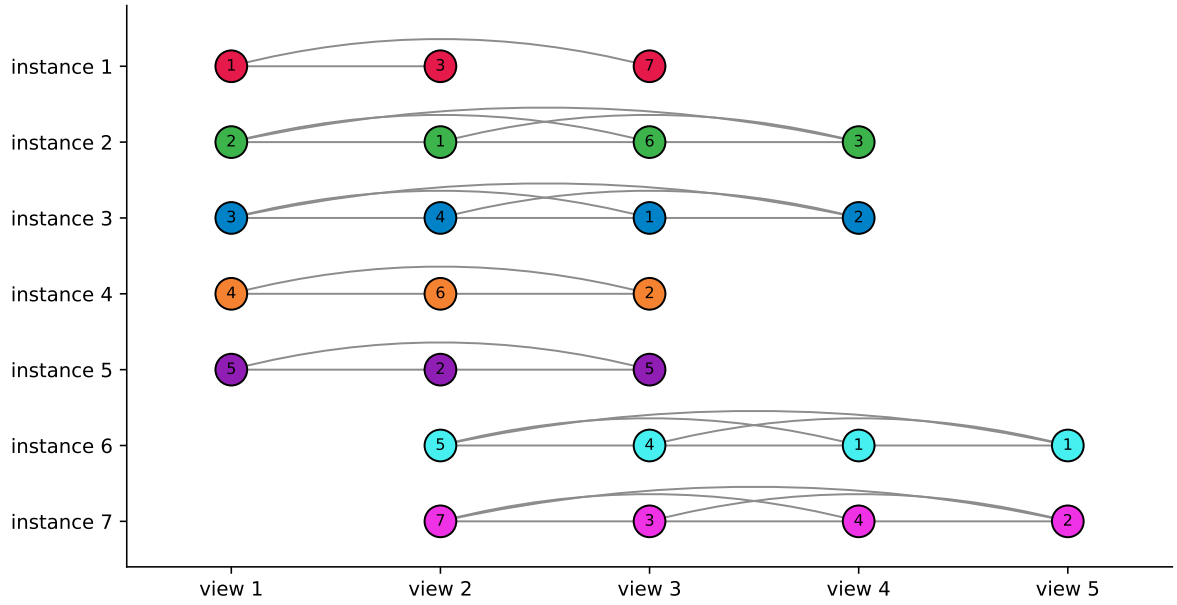


Figure 4.4: Mask graph of the running example. Each column corresponds to a view, each node to an instance mask (labeled with its in-view identifier), and each row to one connected component. The edges represent accepted matches.

For an instance $\mathcal{I}_k$, we aggregate all its mask points

$$\mathcal{P}_k = \{\, P_i(y) \,:\, (i,m) \in \mathcal{I}_k,\ y \in \mathcal{M}_{i,m} \,\}, \tag{4.6}$$

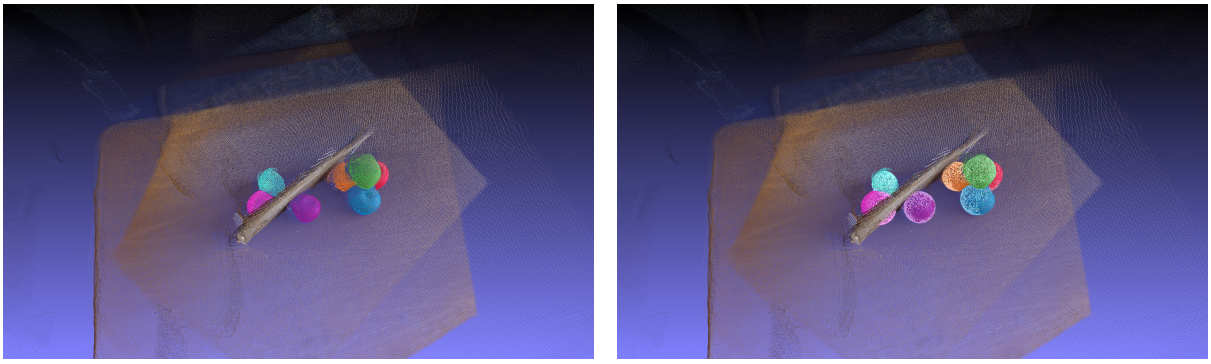
without considering invalid pixels. Each view captures only the side of an apple facing the camera. Thus, the points of an instance cover the part of the apple surface that is visible in at least one of its views, a larger part if the instance consists of masks from multiple views, and only a single cap if it consists of a single mask.

Before fitting, we discard instances supported by fewer than $m_{\min}$ masks. Given enough captured views, we can assume every apple is visible in many views, and so an instance that consists of a single mask is usually a fragment of an apple whose correspondences failed rather than an apple of its own. For sparse captures with a small number of views we set $m_{\min} = 1$, since more apples are likely to be seen in a single view only and should not be discarded for this reason.

We compute the sphere using RANSAC (Section 3.4). A minimal sample consists of four points that determine a unique sphere, provided they do not all lie in the same plane. A point p belongs to the consensus set of a candidate sphere with center c and radius r if its distance to the sphere's surface does not exceed the tolerance δ:

$$\big|\, \|p - c\|_2 - r \,\big| \le \delta. \tag{4.7}$$

After a fixed number of iterations, the sphere with the largest consensus set is chosen, yielding the center point c_k and radius r_k for the instance. We do not refit this sphere based on its consensus set. The consensus criterion makes the fit robust against outliers that remain after filtering in Section 4.2.3. Instances with fewer than four points, for which no sphere can be determined, are discarded.



(a) (b)

Figure 4.5: Sphere fitting in the running example. (a) The seven instances in the reconstructed scene, one color per instance. (b) Their fitted spheres, shown as points. All spheres satisfy the radius criterion.

Using the center of the sphere instead of the centroid of the points as the representative of an instance is essential for the clustering. Since the points often cover only the visible side of the apple, their centroid is shifted toward the observing cameras. Thus, two fragments of the same apple observed from opposite sides would have centroids on opposite sides of the apple, while the centers of their fitted spheres are close to each other.

Finally, we verify the plausibility of the fitted radius. An instance is accepted only if

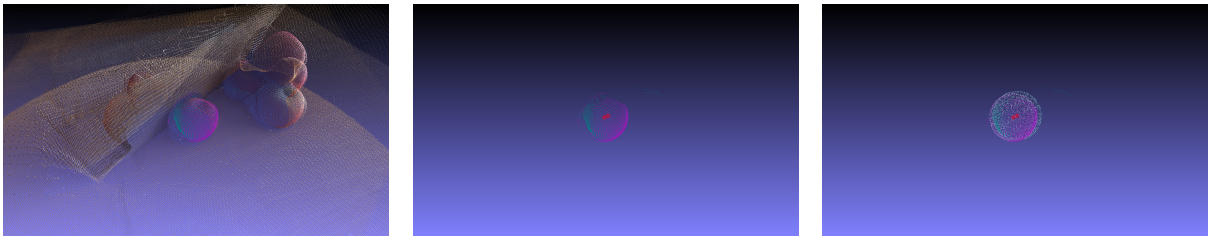
$$r_{\min} \leq r_k \leq r_{\max}, \tag{4.8}$$

and otherwise rejected. For instances that consist of a single mask, we use a looser upper bound, since the points of a single view cover only a cap of the apple and constrain its radius less well. A radius smaller than $r_{\min}$ usually results from small fragments or incorrect masks. A radius exceeding $r_{\max}$ indicates that the instance cannot be a single apple, for example because it contains points from multiple apples or points that do not belong to an apple at all, or because the points from a single view cover too small a portion of the surface to constrain the radius. In the base pipeline, such instances are discarded. The consequences for instances that include multiple apples are analyzed in Section 4.6. Figure 4.5 shows the fitted spheres for the running example.

4.5 Counting: Clustering of Sphere Centers

Since the association links masks exclusively based on correspondences between pairs of views, two instances can describe the same apple even if none of their masks have ever been matched, for example when the views observing them do not share any visible part of the apple's surface, or when SAM3 has split the apple into two masks within a single view. Such split errors result in multiple instances for a single apple, each of which has its own sphere. Since the centers of spheres representing fragments of the same apple are close to one another, these fragments can be merged in 3D space, regardless of the correspondences that failed during the association.

Therefore, we group the centers c_k of all accepted instances using agglomerative clustering with complete linkage (Section 3.5). Two clusters are merged only if all pairs of their centers are closer to



(a) (b) (c)

Figure 4.6: Clustering in the running example, with the correspondences between views 1 and 3 switched off. (a) The two instances of one apple in the reconstructed scene. (b) Their points with the centers of their fitted spheres (red). (c) The fitted spheres nearly coincide, so the clustering merges both instances.

each other than the threshold d_{cl}, so that the diameter of each cluster remains below d_{cl}. Each resulting cluster is considered a single apple, and the number of clusters corresponds to the final count.

The threshold is bounded on both sides. It must be greater than the distance between the centers of the spheres assigned to different fragments of the same apple, so that these fragments are merged. Simultaneously, it must be smaller than the distance between the centers of two adjacent apples, which is approximately the diameter of an apple when the apples are touching, so that neighboring apples are not merged into a single one. The complete linkage ensures that this limit applies to the entire cluster and not just to consecutive pairs of centers. Thus, a row of closely spaced apples cannot be merged into a single cluster, as would be possible when using DBSCAN (Section 3.5).

With the parameters used so far, the running example already recovers every apple as a single instance, so the clustering has nothing to merge. To illustrate the clustering in isolation, Figure 4.6 uses only views 1 and 3 and switches off the correspondences between them ($\tau_{\text{corr}} = 0$), so that every mask forms an instance of its own. The two masks of the same apple thus yield two instances that cover different parts of its surface. The centers of their spheres lie only 0.009 apart, about a fifth of an apple radius and far below $d_{\text{cl}} = 0.040$, so the clustering merges the two instances into one apple.

4.6 Error Analysis of the Base Pipeline

The individual steps of the base pipeline try to correct split and merge errors at different stages, though not to the same extent. This section analyzes where these errors occur and which of them the base pipeline can correct.

Split errors occur whenever the association process fails to combine masks of the same apple. Correspondences between two masks of one apple are missing if both views observe different parts of its surface, if the predictions of the same surface point in both views differ by more than τ_{corr}, or if SAM3 splits the apple into multiple masks within a single view. Matching all pairs of views makes the aggregation robust against individual failed matches (Section 4.3.4), and clustering brings together fragments whose centers are close to each other (Section 4.5). Thus, a split error persists only when the centers of the fragments are farther apart than d_{cl}, which mainly occurs when a fragment consists of a small cap whose sphere is poorly constrained.

Merge errors arise from faulty edges. Correspondences between adjacent apples occur wherever their predicted surfaces come closer than τ_{corr}: at their point of contact and, more generally, wherever

the deviation of the reconstruction is comparable to the distance between the apples. The one-to-one assignment eliminates an incorrect edge as long as the true counterpart of a mask is visible in the other view. If an apple is occluded in view j, its mask in view i has no true counterpart there and is assigned to the mask of a neighboring apple, provided that their overlap exceeds τ_{ov}. Due to the transitivity of connectedness, such an edge connects the complete instances of both apples, and additional false edges can chain multiple apples into a single instance. The same-view constraint blocks such a union only if the two components already contain masks of a common view when the edge is processed.

In the base pipeline, a merged instance usually does not satisfy the radius criterion: its points belong to multiple apples, so the fitted sphere is larger than $r_{\max}$ and the instance is discarded along with all apples it contains. If the merged apples are only partially observed, the fitted radius can remain plausible, and the instance is counted as a single apple. In any case, a merge error leads to undercounting. Unlike a split error, it cannot be corrected by the clustering, which only merges instances and never separates them.

Thus, both types of errors pull the thresholds in opposite directions. A large τ_{corr} and a small τ_{ov} connect the apple fragments even in the case of noisy correspondences, but they also cause incorrect edges between adjacent apples. Strict thresholds prevent false edges, but disrupt the connections within an apple. If the deviation between the predictions for the same surface point from different viewpoints is as large as the distance between neighboring apples, no single choice of thresholds avoids both errors. Furthermore, both errors can cancel each other out in the count, which is why a count close to the actual result does not necessarily indicate a correct association. This motivates the refinement described in the next section: instead of discarding an instance that fails the apple model, it is re-evaluated using stricter thresholds.

4.7 Iterative Refinement

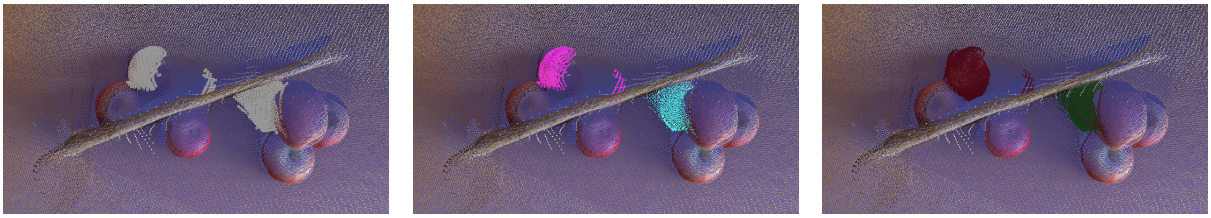
In the refined method, the exclusion of implausible instances is replaced by a re-examination. It follows the base pipeline up to the apple model, but treats an instance that fails the model as a potential merge error and attempts to split it.

An instance is considered too large if its fitted radius exceeds $r_{\max}$ or if the extent of its points exceeds the threshold $e_{\max}$. The radius alone is not sufficient to reliably detect merged instances: since RANSAC selects the sphere with the largest consensus set, it may fit a plausible sphere to the points of a single apple and treat the points of the remaining apples as outliers. Therefore, we additionally measure the extent e_k of an instance, defined as the greatest span of its points along the coordinate axes between the 2nd and 98th percentiles. For an instance whose points belong to multiple apples, the extent exceeds the diameter of a single apple, even if its fitted radius is plausible. As for the radius, instances that consist of a single mask receive a looser bound, since the points of a single view cover only a cap of the apple and are more strongly influenced by the smearing at the mask edge.

An instance that is too large and consists of more than one mask is split by repeating the association (Section 4.3) only for those masks, with stricter thresholds. In iteration t, the thresholds are given by

$$\tau_{\text{corr}}^{(t)} = s^t \, \tau_{\text{corr}}, \qquad \tau_{\text{ov}}^{(t)} = \tau_{\text{ov}} + t \, \Delta, \tag{4.9}$$

with a shrink factor $0 < s < 1$ and an increment $\Delta > 0$. A stricter correspondence threshold excludes matches between neighboring apples whose predicted surfaces are farther apart than two predictions of



(a) (b) (c)

Figure 4.7: Iterative refinement in the running example, with deliberately permissive thresholds. (a) An instance of the initial graph (gray) that contains masks of two apples on both sides of the partition. (b) The stricter association splits it into two instances with one apple each. (c) After clustering, the split-off part of the left apple is merged with the remaining masks of that apple.

the same surface, while a higher overlap threshold additionally rejects weak matches. If the assignment produces multiple components, each of them is compared to the apple model again in the next iteration, and parts that pass the test are accepted, while parts that are still too large are split further. The process ends after at most K iterations.

If a stricter association does not split the instance, or if the maximum number of iterations is reached, the instance is accepted even if it does not match the model. In contrast to the base pipeline, the refined pipeline thus never discards an instance with multiple masks because it is too large, since otherwise every apple it contains would be lost. Single-mask instances that are too large cannot be split and are discarded, as are instances whose radius is smaller than r_{min}. Finally, all instances accepted during the iterations are clustered as before. A split often separates a few masks of an apple from an instance that otherwise belongs to a neighboring apple, and these masks form a fragment that the clustering merges with the remaining masks of their apple.

During splitting, only the masks of the instance itself are considered. Since the masks of other apples are not involved in this association, the one-to-one assignment cannot fall back on the true counterpart of a mask. A mask that was attached to the wrong apple therefore stays with it as long as its overlap with that apple exceeds the stricter threshold.

Figure 4.7 illustrates the refinement on the running example. To induce merge errors, we use deliberately permissive thresholds of $\tau_{\text{corr}} = 0.1$ and $\tau_{\text{ov}} = 0.05$. The original graph again consists of seven components, three of which contain masks of two different apples. Two of these instances exceed r_{max}; the base pipeline discards them and thereby loses two apples completely, resulting in a count of five. The refined pipeline, with $s = 0.25$ and $\Delta = 0.1$, re-associates both instances with $\tau_{\text{corr}}^{(1)} = 0.025$ and $\tau_{\text{ov}}^{(1)} = 0.15$ and splits each of them into two instances, each containing one apple. Subsequently, the clustering process combines the split-off masks with the remaining masks of their respective apples, thereby finding all seven apples. The third mixed instance, consisting of three masks of one apple and one mask of another, satisfies the radius criterion and violates only the extent criterion; however, a stricter association does not split it, so it is accepted as it is. Since the remaining masks of the second apple form an instance of their own, the count remains correct, and five of the seven apples are represented by exactly one instance that contains no other apple, compared to three in the base pipeline.

4.8 Implementation Details

Chapter 5

Results

Describe your qualitative and quantitative evaluations and present your results. Discuss the limitations of your work.

Chapter 6

Conclusions

Give a brief summary of your work and your contributions. Afterwards, briefly outline potential opportunities for future work.

Todo list

data 3
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